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Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr

nums = list(map(int, input().split()))
result = selection_sort(nums)
print(result)
```

INPUT

```
4 2 8 6 1
```

OUTPUT

```
[1, 2, 4, 6, 8]
```

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j >= 0 and arr[j] > key:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key
    return arr
arr = list(map(int, input().split()))
print(insertion_sort(arr))
```

INPUT

```
4 1 3 2
```

OUTPUT

```
[1, 2, 3, 4]
```

Q3. Merge Sort

Score: 10.00 **TC: 3/3 passed**

CODE

```
def merge_sort(arr):
    if len(arr) ≤ 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    return merge(left, right)


def merge(left, right):
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] ≤ right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])

    return result
arr = list(map(int, input().split()))
sorted_arr = merge_sort(arr)
print(sorted_arr)
```

INPUT

```
7 6 5 4
```

OUTPUT

```
[4, 5, 6, 7]
```

Q4. Diagonal Matrix

CODE

```
matrix = eval(input())

is_diagonal = True
n = len(matrix)

for i in range(n):
    for j in range(len(matrix[i])):
        if i ≠ j and matrix[i][j] ≠ 0:
            is_diagonal = False

if is_diagonal:
    print("Diagonal Matrix")
else:
    print("Not Diagonal Matrix")
```

INPUT

```
[[1,0,0],[0,5,0],[0,0,8]]
```

OUTPUT

```
(no output)
```

Q5. Row and Column Sums

— Not Submitted —

Q6. Maximum Subarray Sum

— Not Submitted —

Q7. Common Elements Among Three Lists

— Not Submitted —

Q8. Matrix Multiplication

— Not Submitted —

Q9. Sort Rotate Min Max

— Not Submitted —

Q10. Sort Matrix Search

CODE

```
def sort_matrix_search(matrix, find):
    rows = len(matrix)
    cols = len(matrix[0])
    flat = [num for row in matrix for num in row]
    flat.sort()

    sorted_matrix = []
    for i in range(rows):
        sorted_matrix.append(flat[i*cols:(i+1)*cols])
    position = None
    for i in range(rows):
        for j in range(cols):
            if sorted_matrix[i][j] == find:
                position = (i, j)
                break
        if position:
            break

    if position:
        print("Sorted=", sorted_matrix)
        print("Position=", position)
    else:
        print("Not Found")
matrix = eval(input())
find = int(input())

sort_matrix_search(matrix, find)
```

INPUT

```
[[9,3],[7,1]]
7
```

OUTPUT

```
(no output)
```